

Multiple Choice

1. **Answer:** A

Options: A) reinforcement learning; B) self-supervised learning; C) unsupervised learning; D) supervised learning

Note: Correct option: A - reinforcement learning

2. **Answer:** D

Options: A) To save computing time during testing; B) To save space for storing the Decision Tree; C) To make the training set error smaller; D) To avoid overfitting the training set

Note: Correct option: D - To avoid overfitting the training set

3. **Answer:** B

Options: A) pure; B) not pure; C) useful; D) useless

Note: Correct option: B - not pure

4. **Answer:** A

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: A - True, True

5. **Answer:** C

Options: A) $P(A | B) * P(B | C) * P(C | A)$; B) $P(C | A, B) * P(A) * P(B)$; C) $P(A, B | C) * P(C)$; D) $P(A | B, C) * P(B | A, C) * P(C | A, B)$

Note: Correct option: C - $P(A, B | C) * P(C)$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'False, True'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Decrease variance'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The polynomial degree'.

Long Form Response

11. **Answer:** $\nabla_w f(w) = X^\top Xw - X^\top y + \lambda w$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key